

Task – 1: USB Webcam Installation and Testing on Windows

1. Connect the Webcam

- Insert the USB webcam into any available USB port on the computer.
- If the webcam has a status light, check whether it turns on.

2. Allow Windows to Recognize the Webcam

- Windows normally detects the webcam automatically and installs its basic driver.
- Wait for Windows to display a message indicating that the device is ready.

3. Check the Webcam Driver

- Press **Windows + X** and open **Device Manager**.
- Expand **Cameras** or **Imaging Devices**.
- If a warning symbol appears beside the webcam, right-click it and select **Update driver → Search automatically for drivers**.

4. Launch the Camera Application

- Click the **Start** button and search for **Camera**.
- Open the Camera application.
- Give camera permission if Windows requests access.

5. Verify the Video Output

- Select the connected USB webcam if more than one camera is available.
- Check that the camera displays a clear live preview.

6. Take a Test Photograph

- Sit or stand in front of the webcam.
- Press the **Take Photo** button.
- Open the captured image and confirm that the photograph was saved correctly.

Task – 2: Local Printer Installation and Test Page

1. Connect and Power On the Printer

- For a USB printer, connect its USB cable to the computer.
- For a network printer, connect both the printer and PC to the same Wi-Fi or Ethernet network.
- Switch on the printer and make sure it is in a ready state.

2. Open Printer Settings

- Open **Settings** in Windows.

- Go to **Devices → Printers & scanners**.
- Select **Add a printer or scanner**.

3. Add the Required Printer

- Allow Windows some time to search for available printers.
- Select the required printer and click **Add device**.
- If Windows cannot find it, select **The printer that I want isn't listed** and use the available manual setup method.

4. Install the Printer Driver

- Windows may install the required driver automatically.
- If a driver is required, download the appropriate driver for the printer model and Windows version from the manufacturer's official website.
- Complete the installation and restart the computer if necessary.

5. Verify Printer Status

- Return to **Printers & scanners**.
- Select the installed printer and click **Manage**.
- Make sure its status is displayed as **Ready** or **Online**.

6. Print a Test Page

- Open **Printer properties**.
- Under the **General** tab, select **Print Test Page**.
- Check the printed page to confirm that the printer is functioning properly.

Task – 3: Simulated Printer Connection Troubleshooting

1. Disconnect the Printer

- Switch on the printer.
- Remove the USB cable connecting the printer to the computer.
- Windows may indicate that the printer is disconnected or unavailable.

2. Open Device Manager

- Press **Windows + X** and select **Device Manager**.
- Check sections such as **Printers**, **Print queues**, or **Universal Serial Bus controllers**.

3. Look for Device Errors

- Check whether the printer has a yellow warning symbol.
- Also look for an unknown device or any missing printer entry.

- Such warnings may indicate a driver or hardware-recognition issue.

4. Reconnect the Printer

- Connect the USB cable securely to the computer and printer.
- Try another USB port if the printer is still not detected.

5. Refresh Hardware Detection

- In **Device Manager**, click **Action → Scan for hardware changes**.
- Windows will search for newly connected hardware and update the device list.

6. Update the Printer Driver

- If the printer has a warning symbol, right-click it.
- Select **Update driver → Search automatically for drivers**.
- If required, install the latest compatible driver from the printer manufacturer's website.

7. Confirm the Printer Status

- Open **Settings → Devices → Printers & scanners**.
- Find the printer and check that it is showing **Ready** or **Online**.

8. Perform a Final Test

- Send a test page to the printer.
- If the page prints successfully, the connection problem has been fixed.

Task – 4: Troubleshooting a Webcam Not Working in Zoom

1. Test the Webcam Using Windows Camera

- Open the built-in **Camera** application in Windows.
- Check whether the webcam provides a live video feed.
- If the camera works in this application, the hardware and basic driver are probably working correctly.

2. Close Other Applications

- Exit applications such as **Camera, Microsoft Teams, Skype, and Google Meet**.
- These applications may already be using the webcam.
- Close them completely and open Zoom again.

3. Review Camera Permissions

- Open **Settings → Privacy & security → Camera**.
- Make sure camera access is enabled.

- Turn on **Allow apps to access your camera**.
- Also enable **Allow desktop apps to access your camera**, since Zoom is a desktop application.

4. Choose the Correct Camera in Zoom

- Open **Zoom** and go to **Settings → Video**.
- Under the Camera option, select the connected USB webcam.
- Check the preview window to see whether live video is displayed.

5. Restart the Zoom Application

- Close Zoom completely.
- Start Zoom again and check the camera.
- If the problem continues, restart the computer and try again.

6. Inspect the Webcam Driver

- Press **Windows + X → Device Manager**.
- Expand **Cameras** or **Imaging Devices**.
- Make sure the webcam does not have a yellow warning icon.
- If required, right-click the webcam and choose **Update driver → Search automatically for drivers**.

7. Perform a Final Zoom Test

- Open **Zoom → Settings → Video**.
- Confirm that the webcam preview is visible.
- Start a test meeting to verify that the camera and video function correctly.